Q1

Velocity of a particle of mass 2 kg changes from $\bar{v}_1 = -2\hat{i} - 2\hat{j}$ m/s to $\bar{v}_2 = \hat{i} - \hat{j}$ m/s with a plane friction less surface. Choose the correct option regarding this collision :

(1) The angle made by the plane surface with the positive x-axis is $90^{\circ} + \tan^{-1}\left(\frac{1}{3}\right)$

(2) The angle made by the plane surface with the positive x-axis is $\tan^{-1}\left(\frac{1}{3}\right)$

(3) The direction of change in momentum of the particle makes an angle $\tan^{-1}\left(\frac{1}{3}\right)$ with the positive x-axis.

(4) The direction of change in momentum of the particle makes an angle $90^{\circ} + \tan^{-1}\left(\frac{1}{3}\right)$ with the plane surface.

Q2

A small ball of mass m carrying a charge of Q is dropped in a uniform horizontal magnetic field B. Depth of deepest point of its path from point of release is h. Choose the correct option(s) :

(1) Speed at deepest point is $\sqrt{2gh}$

(2) Speed at deepest point is $\frac{QBh}{m}$

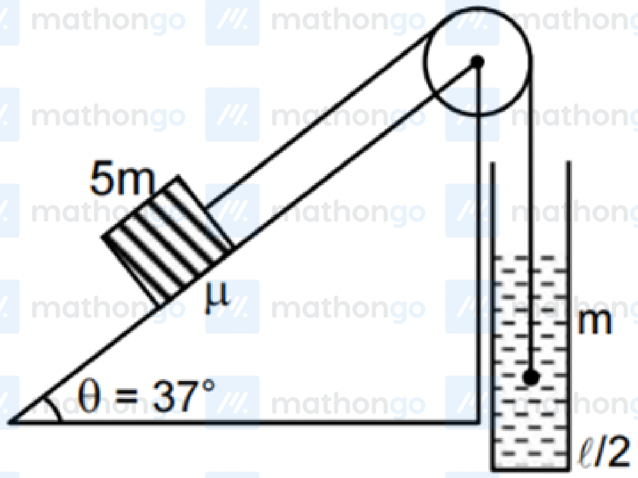
(3) Speed at deepest point is $\frac{2mg}{QB}$

(4) Speed at deepest point is $\frac{2QBh}{m}$

Q3

A block of mass 5 m is connected with a small solid spherical ball of mass m (density $= \rho$) by an ideal string passing over a smooth pulley. The block is on an inclined plane with coefficient of friction $\mu = 0.02$. The spherical ball is in a liquid of density $\frac{\rho}{2}$. The coefficient of viscosity is η. Block is released when string is just taut, then after some time it moves with constant velocity. Assume that the ball always remain inside the fluid.

The radius of ball is r. Choose the correct option(s) :



(1) Acceleration of the ball just after the release $\frac{17g}{60}$

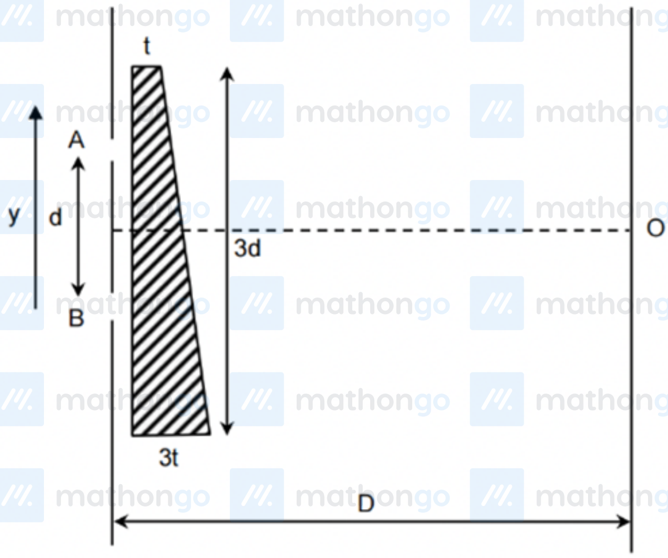
(2) Tension in string just after the release $2.2mg$

(3) Tension in the string when ball move with constant velocity is $2.2mg$

(4) The constant velocity of the ball is $\frac{17mg}{60\pi\eta r}$

Q4

In a standard YDSE two slits are covered by a wedge shape film of refractive index μ as shown in the figure. The thickness of film varies linearly from t to 3 t from one end as shown in the figure. The distance between the two slits is d. The screen is at distance D from slits. White light is used in experiment. Choose the correct option (s) : $(d <<< D)$



Select correct alternative(s)

(1) Central maxima is formed at distance $\frac{2t(\mu-1)D}{3d}$ below O

(2) If the film is displaced $\frac{d}{2}$ along $+y$ axis the central maxima will shift away from O

(3) If the film is displaced $\frac{d}{2}$ along $-y$-axis the central maxima will shift away from O

(4) If the film is displaced $\frac{d}{2}$ along $+y$ axis the fringe width will remain same

Q5

The potential energy U (in joule) of a particle of mass 1 kg moving in $x-y$ plane obeys the law $U = 3x + 4y$, where (x, y) are the co-ordinates of the particle in metre. If the particle is at rest at $(6, 4)$ at time $t = 0$, then:

(1) the particle has constant acceleration

(2) the particle has zero acceleration

(3) the speed of particle when it crosses the y-axis is $10 \mathrm{~m/s}$

(4) coordinates of the particle at $t = 1$ sec are $(4.5, 2)$

Q6

A plano-convex lens is made of a material of refractive index n. When a small object is placed 30 cm away in form of the curved surface of the lens, an image of double the size of the object is produced. Due to reflection from the convex surface of the lens, another faint image is observed at a distance of 10 cm away from the lens. Which of the following statement (s) is(are) true?

(1) The refractive index of the lens is 2.5

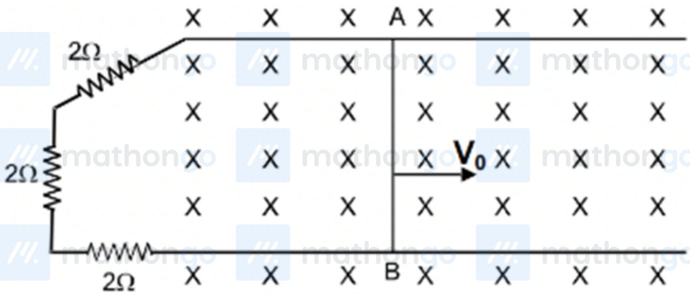
(2) The radius of curvature of the convex surface is 45 cm

(3) The faint image is erect and real

(4) The focal length of the lens is 20 cm

Q7

The conductor AB of mass 1 kg is sliding over two parallel conducting rails separated by a distance of 1 m and is in a region of inward uniform magnetic field $\vec{B}_0 = 0.1(-\hat{k})$. At time $t = 0$, AB is projected towards right with speed v_0.



(1) the velocity of A B as a function of time is given as $v = v_0 e^{-\frac{t}{600}}$

(2) the velocity of rod becomes $\frac{v_0}{2}$ at $t = 600\ell n(2)$

(3) the induced current is $\frac{v_0}{120} A$ at $t = 600\ell n(2)$

(4) the induced emf as a function of time is given as $\left(\frac{v_0}{10}\right) e^{-\frac{t}{600}}$

Q8

The velocity of liquid (v) in steady flow at a location through cylindrical pipe is given by $v = v_0\left(1 - \frac{r^2}{R^2}\right)$, where r is the radial distance of that location from the axis of the pipe and R is the inner radius of pipe. If $R = 10$ cm. volume rate of flow through the pipe is $\pi/2 \times 10^{-2}\ \mathrm{m}^3\ \mathrm{s}^{-1}$ and the coefficient of viscosity of the liquid is $0.75\ \mathrm{Nsm}^{-2}$,

(1) Magnitude of viscous force per unit area at $r = 4$ cm is $6\ \mathrm{N/m}^2$

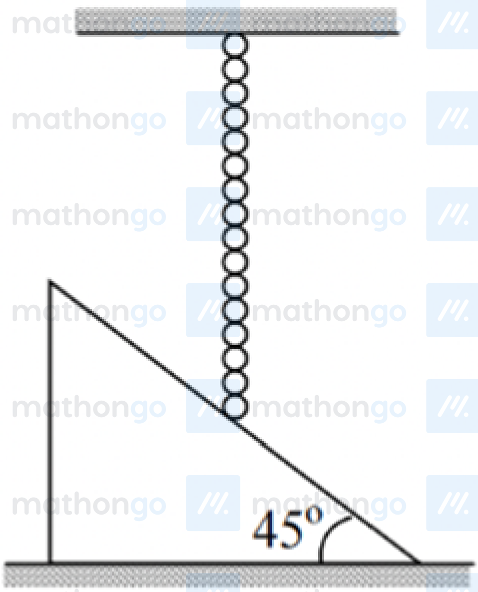
(2) Maximum velocity of flow of liquid is $1 \ \mathrm{m/s}$

(3) Viscous force per unit area increases with radial distance

(4) Viscous force per unit area decreases with radial distance

Q9

Paragraph : A thin flexible uniform chain of mass m and length ℓ is suspended so that its lower end just touches a smooth inelastic plane inclined at 45°. When it is released,



Question :

The force exerted by the chain on the plane at any subsequent instant

(1) $\frac{3mg^2t}{2\sqrt{2}L}$

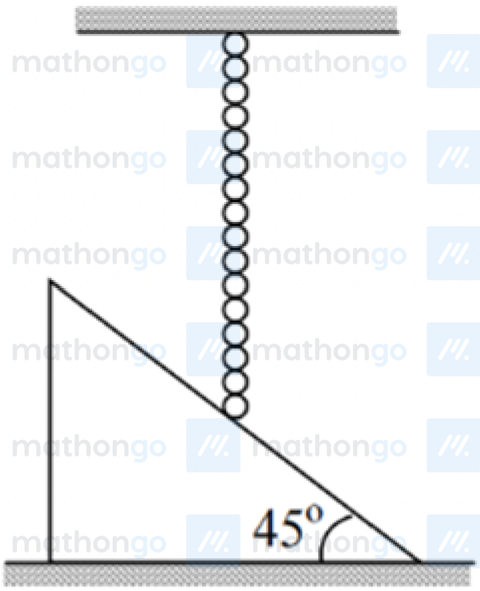
(2) $\frac{3mg^2t^2}{2\sqrt{2}L}$

(3) $\frac{3mgt^2}{2\sqrt{2}L}$

(4) $\frac{mg^2t^2}{2\sqrt{2}\,L}$

Q10

Paragraph : A thin flexible uniform chain of mass m and length ℓ is suspended so that its lower end just touches a smooth inelastic plane inclined at 45°. When it is released,



Question :

The total impulse on the plane while the chain falls

(1) $m(\sqrt{gL})$

(2) $\frac{m}{\sqrt{gL}}$

(3) $m^2(\sqrt{gL})$

(4) $(\sqrt{gL})$

Q11

Parapgrah :

An adiabatic vessel contains $n_1 = 3$ moles of a diatomic gas. Moment of inertia of each molecule is $I = 2.76 \times 10^{-46}$ kg m^2 and root mean square angular velocity is $\omega_0 = 5 \times 10^{12}$rad/sec. Another adiabatic vessel contains $n_2 = 5$ moles of a monoatomic gas at temperature $T_0 = 470$ K. Assuming gases to be ideal both vessels are connected by a thin non-conducting tube. $(\mathrm{K} = 1.38 \times 10^{-23}$ J/molecule$)$

Question : Initial temperature of diatomic gas is

(1) 240 K

(2) 250 K

(3) 220 K

(4) 100 K

Q12

Parapgrah :

An adiabatic vessel contains $n_1 = 3$ moles of a diatomic gas. Moment of inertia of each molecule is $I = 2.76 \times 10^{-46}$ kg m^2 and root mean square angular velocity is $\omega_0 = 5 \times 10^{12} \mathrm{rad/sec}$. Another adiabatic vessel contains $n_2 = 5$ moles of a monoatomic gas at temperature $T_0 = 470$ K. Assuming gases to be ideal both vessels are connected by a thin non-conducting tube. $\left(\mathrm{K} = 1.38 \times 10^{-23}\ \mathrm{J/molecule}\right)$

Question : Root mean square angular velocity of diatomic molecules when both vessels are connected

(1) $3 \times 10^{12} \mathrm{rad/sec}$

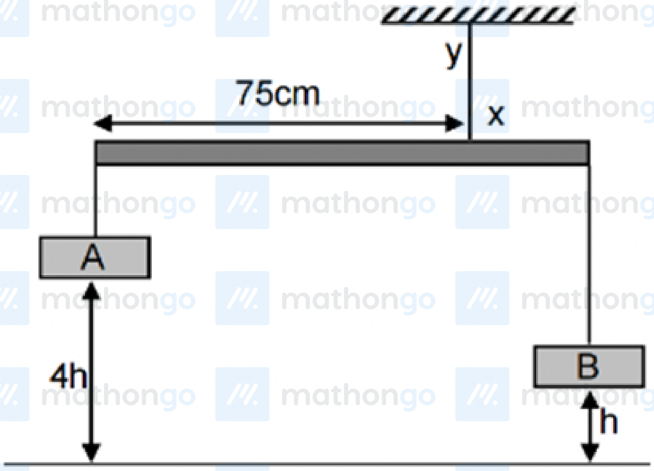
(2) $1 \times 10^{10} \mathrm{rad/sec}$

(3) $5 \times 10^{13} \mathrm{rad/sec}$

(4) $6 \times 10^{12} \mathrm{rad/sec}$

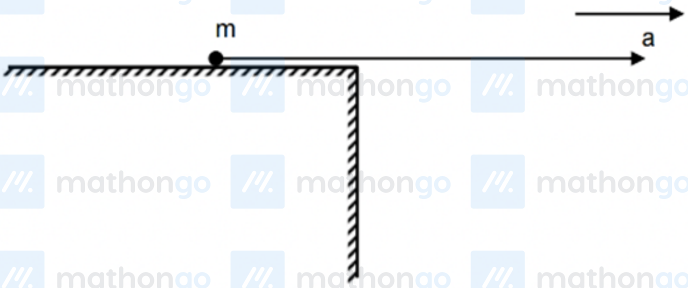
Q13

Horizontal rod of length 1 m given below is in equilibrium. If the potential energies of objects A and B are equal, find the tension (in N) in string xy. (Rod is homogeneous and weight of it is 1 N)



Q14

A small particle is attached to an inextensible string and placed on a table. The other end is pulled with constant acceleration $\frac{4g}{3}$ towards right. Initially the particle was at rest. After the particle leaves the edge of table, the maximum vertical displacement of the particle is $\frac{48}{25}$ m. Find the length of string in m.



Q15

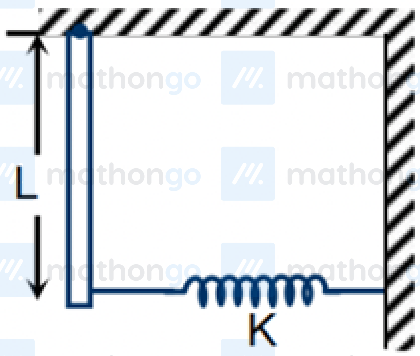
A thermally insulated piece of metal is heated under atmospheric pressure by an electric current so that it receives electric energy at a constant power P. This leads to an increase of the absolute temperature T of the metal with time t as given by the equation $T(t)=T_0[1+a(t-t_0)]^{1/4}$. Here a, t_0 and T_0 are constants. The heat capacity of the metal is $C_P(T)=\frac{4P}{a_0^4}T^{\mathrm{n}}$ (which is temperature dependent in the temperature range of the experiment), then find the value of ' n '.

Q16

An earth satellite is revolving in a circular orbit of radius a with a velocity v_0. A gun in the satellite is directly aimed toward earth. A bullet is fired from the gun with muzzle velocity $\frac{v_0}{2}$. Find the ratio of distance of farthest and closest approach of bullet from centre of earth. (Assume that mass of the satellite is very-very large with respect to the mass of the bullet)

Q17

A straight rod of length L and mass m is pivoted freely from a point of the roof. Its lower end is connected to an ideal spring of spring constant K. The other end of the spring is connected to the wall as shown in the figure. The frequency of small oscillation of the system is found to be $\frac{1}{2\pi}\sqrt{r\left(\frac{K}{m}+\frac{g}{2L}\right)}$, find r



Q18

A source of sound of frequency 1.8kHz moves uniformly along a straight line at a distance 250 m from observer. The velocity of source is 0.8C where C is the velocity of sound. Find out the frequency of sound received by observer (in kHz) at the moment when the source gets closest to him.

Q19

A is a black compound of manganese. It reacts with a halogen acid B to give a greenish yellow gas C. When excess of C reacts with NH_3, an unstable trihalide is formed and B is obtained back.

(1) The halogen acid B is HF.

(2) Gas C is F_2.

(3) Hybridization of N remains same in the product as in NH_3.

(4) When excess of NH_3 reacts with C, N_2 is formed.

Q20

The most reactive halogen reacts with other halogens Cl_2, Br_2 and I_2 under controlled conditions forming tetra-atomic, hexa-atomic and octa-atomic inter-halogen compounds respectively. Select the incorrect statement(s):

(1) The octa-atomic compound contains no lone pair.

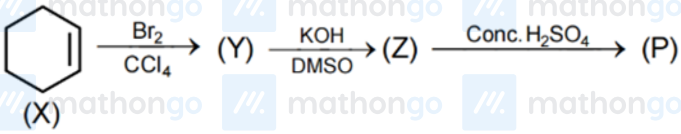
(2) Hybridization of central atom in the hexa-atomic molecule is $sp^3\ d^2$.

(3) The tetra-atomic compound is planar.

(4) One of the tetra-atomic compound is used in enrichment of ^{235}U

Q21

Which of the following is/are true for products of the following reaction sequence?



(1) Y is formed by electrophilic addition on X

(2) Z is formed by nucleophilic substitution of Y

(3) Formation of P from Z involves rearrangement

(4) P and X are identical

Q22

Select the correct option(s) about the aqueous solution that contains $3.42 ppm (w/v) Al_2(SO_4)_3$ and $1.42 ppm (w/v) Na_2SO_4$:

(1) $\left[Al^{3+}\right] = [Na^+]$

(2) $\left[SO_4^{2-}\right] = [Na^+] = \left[Al^{3+}\right]$

(3) $\left[SO_4^{2-}\right] = [Na^+] + \left[Al^{3+}\right]$

(4) $\left[SO_4^{2-}\right] = [Na^+]$

Q23

The complex $\left[Fe(H_2O)_5NO\right]^{2+}$ is formed in the NO_3^- radical brown ring test. Select the correct statement(s) regarding this complex :

(1) Colour is due to charge transfer.

(2) It has iron in $+1$ oxidation state and NO^+ ligand.

(3) It has magnetic moment of 3.87BM with three unpaired electrons.

(4) It has d^2sp^3 hybridisation.

Q24

α-maltose$(C_{12}H_{22}O_{11})$ can be hydrolysed to glucose $(C_6H_{12}O_6)$ according to the following reaction:

$C_{12}H_{22}O_{11}(aq) + H_2O(\ell) \rightarrow 2C_6H_{12}O_6(aq)$

Given:

Standard enthalpy of formation of $C_{12}H_{22}O_{11}(aq) = -2238$ kJ/mol

Standard enthalpy of formation of $H_2O(\ell) = -285$ kJ/mol

Standard enthalpy of formation of $C_6H_{12}O_6(aq) = -1263$ kJ/mol

Time (min.)	0	50	100
Conc. of α-maltose (M)	4.0	1.0	0.25

Which of the following statement(s) is/are true?

(1) The hydrolysis of α-maltose is exothermic.

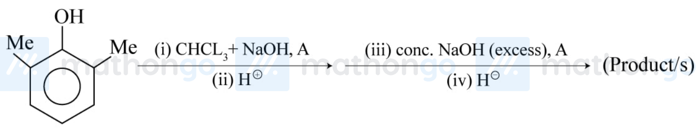
(2) Heat liberated in combustion of 1.0 mol of α-maltose is greater than the heat liberated in combustion of 2.0 mol of glucose.

(3) Increasing temperature will increase the degree of hydrolysis of α-maltose.

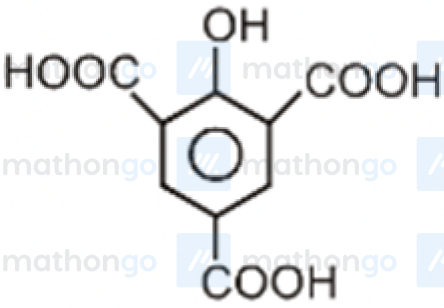
(4) The hydrolysis of α-maltose follow 1^{st} order kinetics.

Q25

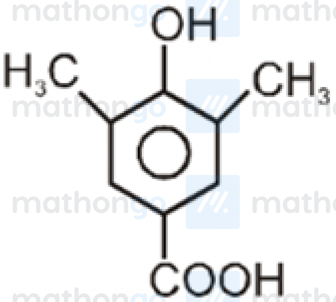
The product/s of the following reaction is/are :



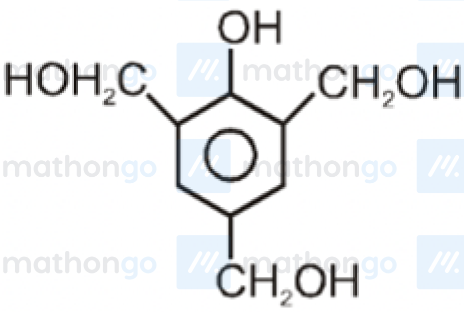
(1)



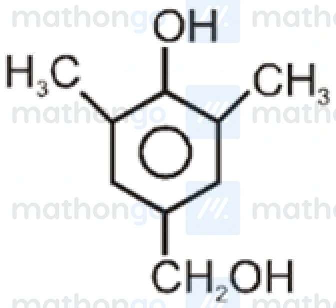
(2)



(3)

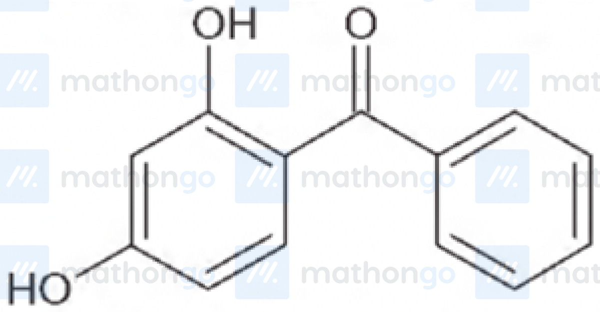


(4)



Q26

A sunburn prevention agent must be non-toxic, non-staining, non-volatile, reasonably water soluble. It should filter of the burning rays of sunlight. Given structure is one such compound:



which method/s for synthesis of above compound starting from m-dimethoxybenzene:

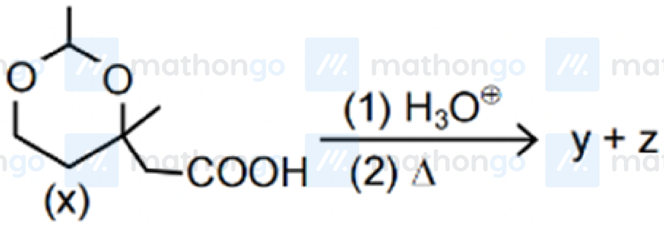
(1) (i) Ph – COCl, (ii) dil. NaOH

(2) (i) Conc. HI (ii) Ph – COCl, $AlCl_3$

(3) (i) Ph – COCl, $AlCl_3$ (ii) conc. HI

(4) All of these

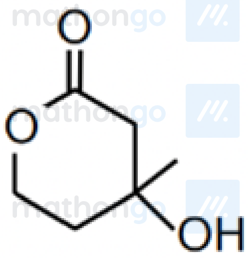
Q27

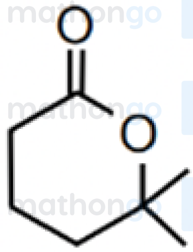


Paragraph :

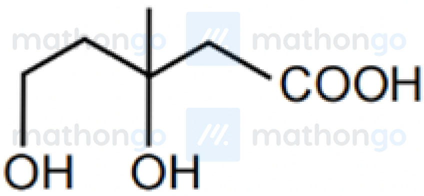
Test	Compound y	Compound z
(1) NaOI	No reaction	Yellow ppt
(2) $[Ag(NH_3)_2]^{\oplus}$	No reaction	Silver mirror
(3) $NaHCO_3$	No reaction	No reaction

Question : Compound y is :

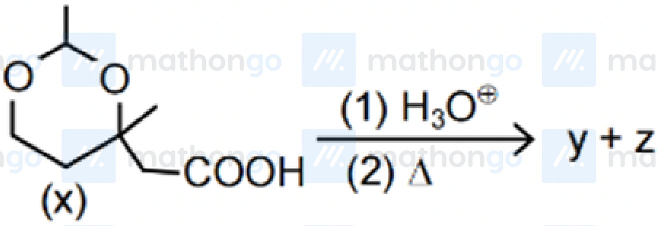
(1) 

(2) 

(3) $CH_3 - CH = O$

(4) 

Q28



Paragraph :

Test	Compound y	Compound z
(1) NaOI	No reaction	Yellow ppt
(2) $\left[Ag(NH_3)_2\right]^{\oplus}$	No reaction	Silver mirror
(3) $NaHCO_3$	No reaction	No reaction

Question : Choose the correct option :

(1) Compound z is $CH_3 - CH = O$

(2) Compound y is formed due to decarboxylation

(3) Compound y is formed due to esterification between $-COOH$ & $-$ OH group

(4) Both A \& C are correct.

Q29

Paragraph :100 mL, 0.1M weak acid HA $\left(K_a = 10^{-6}\right)$ is titrated with VmL, 0.1MNaOH at 25° C.

[Given : $\log 3 = 0.48, \log 2 = 0.30$]

Question : The pH of final solution when $V = 25$ mL is :

(1) 5.52

(2) 6.48

(3) 8.48

(4) 7.52

Q30

Paragraph :100 mL, 0.1M weak acid HA $\left(K_a = 10^{-6}\right)$ is titrated with VmL, 0.1MNaOH at 25° C.

[Given : $\log 3 = 0.48, \log 2 = 0.30$]

Question : The pH of final solution when $V = 100$ mL is :

(1) 4.65

(2) 3.35

(3) 9.35

(4) 10.65

Q31

A 0.25 molal aqueous solution of $Co(NH_3)_4Cl_3$ freezes at -0.93° C. Calculate the number of stereoisomers for this coordination compound. Molal depression constant for H_2O is 1.86 K/molal. (Assume complete dissociation of the coordination compound into its ions)

Q32

How many of the following contain peroxide linkage?

(a) $[FeO_4]^{2-}$

(b) Mn_2O_7

(c) HSO_5^-

(d) CrO_5

(e) PbO_3^{2-}

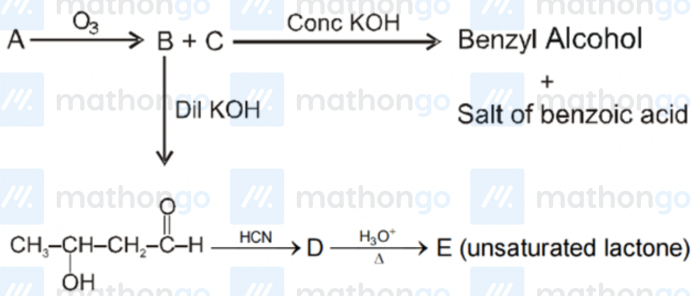
(f) HNO_4

(g) $Na_2\ B_4O_7$

(h) MnO_2

(i) $S_4O_6^{2-}$

Q33



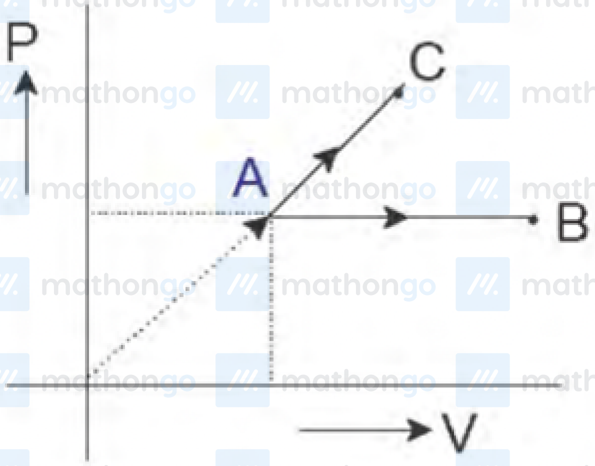
Total number of carbon atoms in E = answer

Q34

Write the total number of cyclic isomers of C_5H_{10} (consider stereoisomers also)

Q35

One mole ideal monoatomic gas is heated according to path AB and AC. If temperature of state B and state C are equal. Calculate $\frac{q_{AC}}{q_{AB}} \times 10$.



Q36

A volume of 100ml of a liquid contained in an isolated container at a pressure of 1bar. The pressure is steeply increases to 100 bar by which the volume of liquid is decreased by 1ml. The change in enthalpy, ΔH, of the liquid (in bar mL)

Q37

Let $p(z) = z^3 + az^2 + bz + c$ where $a,\ b,\ c$ are real. There exists a complex number ω such that three roots of $p(z) = 0$ are $\omega + 3i, \omega + 9i, 2\omega - 4$ where $i^2 = -1$, then

(1) 4 is real root of $p(z) = 0$

(2) $a = -12, b = 84, c = 208$

(3) $|a + b + c| = 136$

(4) $|a + b + c| = 280$

Q38

If n is the smallest positive integer such that the polynomial $x^4 - n \cdot x + 63$ can be written as a product of a linear and a cubic polynomial with integer coefficients, then

(1) Number of divisors of n is 10

(2) Number of natural solutions of $x.\,y = n$ is 10

(3) Number of odd divisors of n is 2

(4) $\int_0^n \frac{e^x}{e^{[x]}} dx = 100(e-1)$ [where [.] denotes greatest integer function]

Q39

A circle with centre O and radius R is drawn on a paper and A is a given point inside circle with $\mathrm{OA} = \mathrm{a}$ where O is origin and A is on positive x axis. Fold the paper to make a point A' on the circumference coincident with point A, then a crease line is left on the paper. Set of all points on such crease lines, when A' goes through every point on the circumference

(1) consist of all points on the border of or outside the ellipse with centre $\left(\frac{\mathrm{a}}{2}, 0\right)$

(2) consist of all points on the border of or outside the ellipse having eccentricity $\frac{a}{R}$

(3) Consist of all points on the border of or inside the ellipse having eccentricity $\frac{a}{R}$

(4) Consist of all points on the border of or inside the ellipse where principal axes are parallel to co-ordinate axes.

Q40

Let $f: R \rightarrow R$ be a function such that $f(xy+1) = f(x) \cdot f(y) - f(y) - x - 2$ holds $\forall x, y \in R$ if $f(0) = 1$ then

(1) $f(x) = x^2 + 1$

(2) $\sin^{-1} \cdot \left(\sin\left(f^{-1}(7)\right)\right) = 6 - 2\pi$

(3) $\sum_{\mathrm{k}=0}^{10} \mathrm{f}(\mathrm{k}) = 66$

(4) $f(2016)$ is prime

Q41

Let A is a square matrix of order 2 and e^A is defined as

$\mathrm{e}^{\mathrm{A}} = \mathrm{I} + \mathrm{A} + \frac{\mathrm{A}^2}{2!} + \frac{A^3}{3!} + \ldots\ldots\ldots = \frac{1}{2}\begin{bmatrix} f(x) & g(x) \\ g(x) & f(x) \end{bmatrix}$ where $A = \begin{bmatrix} x & x \\ x & x \end{bmatrix}$; $0 < x < 1$ where I is a identity matrix.

(1) If $\int_0^1 \left(f\left(\frac{x^2}{2}\right) - 1\right)(x-\alpha) dx = 0$ then $1 < \alpha < 2$

(2) If $\int_0^1 \left(f\left(\frac{x^2}{2}\right) - 1\right)(x-\alpha)dx = 0$ then $0 < \alpha < 1$

(3) General solution of differential equation $f(x) \cdot dy + yg(x)dx = 0$ is $y = \frac{c \cdot e^x}{f(x)}$ where c is arbitrary constant.

(4) General solution of differential equation $f(x) \cdot dy + y \cdot g(x)dx = 0$ is $y = c.\, e^x g(x)$ where c is arbitrary constant

Q42

If expression $1+x, 1+x+x^2, 1+x+x^2+x^3, \ldots\ldots.. 1+x+x^2 \ldots\ldots + x^n$ are multiplied together and terms of the product are arranged in the increasing powers of x in the form $a_0 + a_1 x + a_2 x^2 + \ldots\ldots$ then

(1) Number of terms in the product is $\frac{(n+1)(n+2)}{2}$

(2) Number of terms in the product is $\frac{n^2+n+2}{2}$

(3) Coefficients of the terms equidistant from beginning and end are equal

(4) $a_1 + a_3 + a_5 + \ldots\ldots\ldots = a_0 + a_2 + a_4 + \ldots\ldots\ldots = (n+1)!$

Q43

Let P, Q and R be three non- singular matrices of order 3 that $P^2 + 2I = 0, \det(2R - P^2) = 32$ and satisfying $P^5 - 2P^3R + QP^2 - 2QR = 0$, Then $|Q-P|^2$ is Not equal to

(1) -8

(2) 8

(3) 12

(4) 16

Q44

Let f : $[0,1] \rightarrow R$ be a continuous function, $g(x)$ be a non-increasing function on [0,1] and $h(x) : [0,\infty) \rightarrow [0,\infty)$ be a continuous strictly increasing function with $h(0) = 0$, then

(1) $\alpha \int_0^1 g(x)dx \leq \int_0^\alpha g(x)dx$ for any $\alpha \in (0,1)$

(2) $\left(\int_0^1 f(t)dt\right)^2 \leq \int_0^1 f^2(t)dt$

(3) $\int_0^a h(x)dx + \int_0^b h^{-1}(x)dx \geq ab$

(4) $\int_0^a h(x)dx + \int_0^b h^{-1}(x)dx \leq ab$

Q45

Paragraph : We know that in a chess board, there are $8 \times 8 = 64$ small squares. Now answer the following questions based on a chess board 5.

Question : If two of the 64 squares are chosen at random then the probability that they don't have a side in common is-

(1) $\frac{1}{9}$

(2) $\frac{8}{9}$

(3) $\frac{1}{18}$

(4) $\frac{17}{18}$

Q46

Paragraph : We know that in a chess board, there are $8 \times 8 = 64$ small squares. Now answer the following questions based on a chess board 5.

Question : If three small squares are chosen at random on a chess board, probability that they are in a diagonal line, is-

(1) $\frac{9}{744}$

(2) $\frac{13}{744}$

(3) $\frac{7}{744}$

(4) $\frac{11}{744}$

Q47

Paragraph :A real valued continuous function $y = f(x)$ is increasing or decreasing in an interval iff $\frac{dy}{dx} > 0$ or $\frac{dy}{dx} < 0$ respective also if f is continuous in [a, b] differentiable in (a, b) and $f(a) = f(b)$, then the equation

$f'(x)=0$ has at least one value of x (i.e. root of $f'(x)=0$) in (a,b) Based upon the above information answer the following question.

Question : Let $F(x)=\int_{\sin x}^{\cos x} e^{\left(1+\sin^{-1} t\right)^2} dt$ for $x \in \left[0, \frac{\pi}{2}\right]$, Then

(1) There exists some $\mathrm{c} \in \left(0, \frac{\pi}{2}\right)$ such that $\mathrm{F}'(\mathrm{c})=0$

(2) There exists some $\mathrm{c} \in \left(0, \frac{\pi}{2}\right)$ such that $\mathrm{F}''(\mathrm{c})=0$

(3) There exists some $c \in \left(0, \frac{\pi}{2}\right)$ such that $F'''(c)=0$

(4) There is exactly one value of $c \in \left(0, \frac{\pi}{2}\right)$ such that $F'(c)=0$

Q48

Paragraph :A real valued continuous function $y=f(x)$ is increasing or decreasing in an interval iff $\frac{dy}{dx}>0$ or $\frac{dy}{dx}<0$ respective also if f is continuous in [a, b] differentiable in (a,b) and $f(a)=f(b)$, then the equation $f'(x)=0$ has at least one value of x (i.e. root of $f'(x)=0$) in (a,b) Based upon the above information answer the following question.

Question : If a function $y=f(x)$ is such that $f'(x)$ is continuous function and satisfies $(f(x))^2=m^2+\int_0^x\left((f(t))^2+(f'(t))^2\right)dt, m \in R^+$, Then

(1) f is increasing for all $x \in R$

(2) f is bounded function

(3) f is even function

(4) if $m=100$, Then $f(0)=\frac{m}{10}$

Q49

In a ΔABC If medians from B and C are perpendicular and minimum value of $\cot B+\cot C$ is k then find the value of 6k.

Q50

Let O be an interior point of ΔABC such that $\overrightarrow{\mathrm{OA}}+2\overrightarrow{\mathrm{OB}}+3\overrightarrow{\mathrm{OC}}=\overrightarrow{0}$. If area of $\Delta ABC=k$ (Area of ΔAOC) then find k.

Q51

If m be the number of integers n satisfying $[\sqrt{n}]=[\sqrt{n+125}]$ and $1 \leq n \leq 5000$ (where [.] denotes greatest integer function) then find sum of digits of m.

Q52

Find number of all 4-tupple (a, b, c, d) of natural numbers with $a \leq b \leq c$ and $a!+b!+c!=3^{d}$

Q53

Let a line with the inclination angle of 60° be drawn through the focus F of the parabola $y^{2}=8(x+2)$. If the two intersection points of line and parabola are A and B and the perpendicular bisector of the chord AB intersects the x - axis at point P. If the length of bisector of the segment PF is ℓ, then $\frac{3}{2} \ell$ is equal to

Q54

TP and TQ are any two tangents to a parabola and the tangent at a third point R cuts them is P' and Q', then find $\frac{T P'}{T P}+\frac{T Q'}{T Q}$

Answer Key

Q1 (1) (3)	**Q2** (1) (2) (3)	**Q3** (1) (3) (4)	**Q4** (1) (4)
Q5 (1) (3) (4)	**Q6** (1) (4)	**Q7** (1) (2) (3) (4)	**Q8** (1) (2) (3)
Q9 (2)	**Q10** (1)	**Q11** (2)	**Q12** (4)
Q13 (6)	**Q14** (2)	**Q15** (3)	**Q16** (3)
Q17 (3)	**Q18** (5)	**Q19** (3) (4)	**Q20** (1)
Q21 (1) (2) (3)	**Q22** (1) (3)	**Q23** (1) (2) (3)	**Q24** (1) (2) (4)
Q25 (2) (4)	**Q26** (2) (3)	**Q27** (1)	**Q28** (4)
Q29 (1)	**Q30** (3)	**Q31** (2)	**Q32** (3)
Q33 (5)	**Q34** (7)	**Q35** (8)	**Q36** (9900)
Q37 (1) (3)	**Q38** (1) (2) (3)	**Q39** (1) (2)	**Q40** (2) (3) (4)
Q41 (2) (3)	**Q42** (2) (3)	**Q43** (2) (3) (4)	**Q44** (1) (2) (3)
Q45 (4)	**Q46** (3)	**Q47** (2)	**Q48** (1)
Q49 (4)	**Q50** (3)	**Q51** (9)	**Q52** (3)
Q53 (8)	**Q54** (1)		

Q1

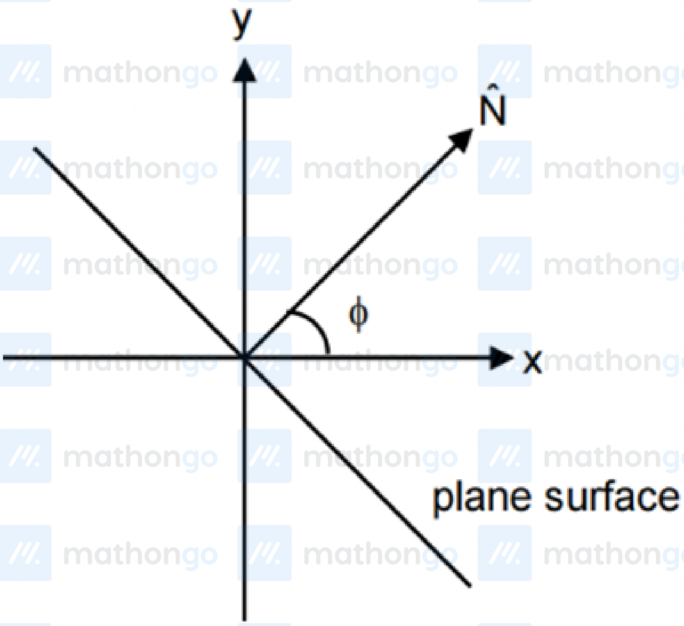
$\widehat{N} = \frac{\vec{v}_2 - \vec{v}_1}{\left|\vec{v}_2 - \vec{v}_1\right|} = \frac{3\hat{i}+\hat{j}}{\sqrt{10}}$

angle $b/w \hat{N}$ and x-axis

$\cos\phi = \hat{i}. \quad \widehat{N} = \frac{3}{\sqrt{10}}$

$\tan\phi = \frac{1}{3}$

Impulse is change in momentum. Hence, impulse = 2 (v2 - v1) = 2 (3i + j)



Angle b/w plane surface and $+x$ axis $= 90 + \phi$.

Q2

The force acting on a particle of charge Q that moves with a velocity of v in a magnetic field is given by the formula:

$\vec{F} = \vec{Q}v \times \vec{B}$

This force is perpendicular to both the velocity and the magnetic field, so as in the present case the magnetic field is uniform and horizontal, the ball will move along a curve in a vertical plane perpendicular to the field lines.

Conservation of energy gives $v = \sqrt{2gh}$.

$Qv_yB = ma_x. \quad \Rightarrow QB \times \frac{dy}{dt} = m \times \frac{dv_x}{dt}$

$QBh = mv = m\sqrt{2gh}$

By substitution we can get rest of the answers

Q3

Just after release

$5mg\sin 37^{\circ} - 5\mu mg\cos 37^{\circ} - mg_{eff.} = 6ma$

$g_{\text{eff}} = g\left(1 - \frac{\rho}{2\rho}\right) = \frac{g}{2}$

$5mg\frac{3}{5} - 5m(0.2)g\frac{4}{5} - \frac{mg}{2} = 6ma$

$a = \frac{17g}{60}$

For terminal velocity

$1.7mg = 6\pi\eta rv$

$v = \frac{17mg}{60\pi\eta r}$

Q4

Shift in fringe pattern = distance of central maxima

$= \frac{t_2(\mu-1)}{\lambda}\beta - \frac{t_1(\mu-1)}{\lambda}\beta = \frac{(t_2-t_1)(\mu-1)\beta}{\lambda}$

$= \frac{d\tan\theta(\mu-1)\beta}{\lambda}(\text{constant}) = \frac{2t(\mu-1)D}{3d}$

Shift in fringe pattern is constant. Hence, if film is displaced d/2 along +y axis the fringe width will remain same.

Q5

$\vec{a} = \frac{\vec{F}}{m}$

$\because\ m = 1\text{ kg}$

$\vec{a} = -\left(\frac{\partial U}{dx}\hat{i} + \frac{\partial U}{dx}\hat{j}\right) = -3\hat{i} - 4\hat{j}$

$\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2 \Rightarrow (x-6) = \frac{1}{2}\times(-3)\times t^2 \Rightarrow x = 6 - \frac{3t^2}{2}$

and $y - 4 = \frac{1}{2}\times(-4)\times t^2 \Rightarrow y = 4 - 2t^2$

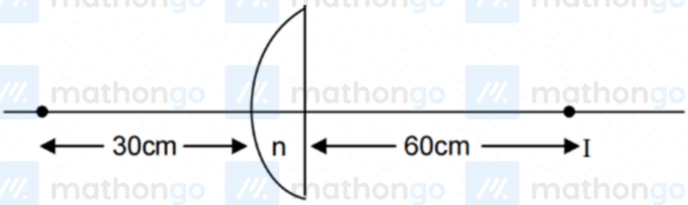
So, at $t = 1, x = 4.5, y = 2$

When particle crosses y-axis, $x=0 \Rightarrow t=2$

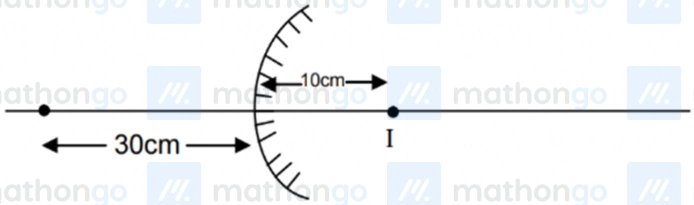
At $t=2, \ |\vec{v}|=|\vec{u}+(\vec{a})(2)|=\left|-6\hat{i}-8\hat{j}\right|=10$ m/s

Q6

Case-I



$\frac{1}{60}+\frac{1}{30}=\frac{1}{f_1}$
$\Rightarrow \frac{1}{f_1}=\frac{1}{60}+\frac{2}{60}$
$f_1=\frac{R}{n-1}=+20$ cm



Case-2

$\frac{1}{10}-\frac{1}{30}=\frac{1}{f_2}$
$\frac{3}{30}-\frac{1}{30}=\frac{1}{f_2}=\frac{2}{30}$
$f_2=15=\frac{R}{2}\Rightarrow R=30$
$=2\text{n}-2=3$

$\Rightarrow f_1=+20\ cm$

Refracted index of lens is 2.5

ROC of convex surface is 30 cm

Faint image is erect and virtual

focal length of lens is 20 cm

Q7

Retarding force $F = B_0 I \ell$

$= B_0\left(\frac{B_0 v \ell}{R}\right)\ell = \frac{B_0^2 \cdot v}{6}$

Retardation$= \frac{B_0^2 v}{6\times1}$ (at any instant of time)($\because m = 1$)

$\frac{B_0^2 v}{6} = -\frac{dv}{dt} \Rightarrow \int_{v_0}^{v}\frac{-dv}{v} = \int_0^t\left(\frac{B_0^2}{6}\right)dt$

$\Rightarrow \ell n\left[\frac{v_0}{v}\right] = \frac{t}{600}$

$v = v_0 e^{-\frac{t}{600}}$

At $t = 600\ell n(2)$

$\Rightarrow \quad v = \frac{v_0}{2}$

$i = \frac{B_0 v\ell}{R} = \frac{B_0 v}{R} = \frac{B_0 v_0}{12} = \frac{v_0}{120}$

$[\because R = 6\Omega, |B_0| = 0.1\ T]$

$E = B_0 v \ell = \frac{v_0}{10}e^{-\frac{t}{600}}$

Q8

Magnitude of viscous force, $F = \eta A\frac{dv}{dr}$

$\Rightarrow$ viscous force per unit area $\sigma = \frac{F}{A} = \eta\frac{dv}{dr}$

$v = v_0\left(1 - \frac{r^2}{R^2}\right) \Rightarrow \frac{dv}{dr} = -\frac{2V_0 r}{R^2}\sigma = \eta\cdot\frac{2v_0 r}{R^2}$

Volume rate of flow, Q consider an annular element at r from axis, width dr.

$dA = 2\pi r dr; dQ = v\cdot dA = v_0\left(1 - \frac{r^2}{R^2}\right)2\pi r dr$

Q9

$Q = \int dQ = 2\pi v_0\left[\frac{r^2}{2} - \frac{r^4}{4R^2}\right]_0^R = \frac{\pi}{2}R^2 v_0$

$\Rightarrow v_0 = \frac{2Q}{\pi R^2}$

$\therefore \quad$ (i) $\Rightarrow \quad \sigma = \eta\frac{4Q}{\pi R^4}r, R = 0.1\text{ m}$

At $r = 0.04m, \sigma = (0.75)\ 4\times\frac{\pi}{2}\times10^{-2}\times\frac{0.04}{\pi\times10^{-4}} = 6\text{Nm}^{-2}$

When the chain is released all its links are in free fall, so the chain remains vertical but without tension. After time$t$, the top link has fallen a distance $\frac{1}{2}gt^2$.

Normal force on the plane exerted due to weight, $N = \frac{mg^2t^2}{2L}\cos 45^\circ = \frac{mg^2t^2}{2\sqrt{2}L}$

In addition to weight the impulsive force due to change in momentum also acts. During a small interval of time Δt, a length gt Δt of mass $\frac{mgt\Delta t}{L}$ has the component of its velocity perpendicular to the plane reduced from $\frac{gt}{\sqrt{2}}$ to zero. From impulse momentum equation we have

$F.\Delta t=\frac{mgt\ \Delta t}{L}\left(0-\frac{gt}{\sqrt{2}}\right)$

Or $F=-\frac{mg^2t^2}{L\sqrt{2}}$

Minus sign indicates that the force on the chain is opposite to its velocity.

The total normal force at any instant, while the chain is falling, is $F_N=\frac{mg^2t^2}{2L\sqrt{2}}+\frac{mg^2t^2}{L\sqrt{2}}=\frac{3mg^2t^2}{2\sqrt{2}L}$

Q10

The time of fall, $t=\sqrt{\frac{2L}{g}}$

The total impulse $=\int_0^{\sqrt{2L/g}}F_N dt$

$=\int_0^{\sqrt{2L/g}}\frac{3mg^2t^2}{2\sqrt{2}\,\mathrm{L}}dt$

$=\frac{3mg^2}{2\sqrt{2}L}\left[\frac{t^3}{3}\right]_0^{\sqrt{2L/g}}$

$=m(\sqrt{gL})$

Q11

Energy associated with each degree of freedom per molecule $=\frac{1}{2}KT$

So, for diatomic molecules, energy associated with rotational degree of freedom $=2\times\frac{1}{2}KT$

(Since there are two rotational degree of freedom and ω_0 is the net R.M.S. angular velocity)

$\Rightarrow 2\times\frac{1}{2}KT=\frac{1}{2}\times I\omega^2$

$\Rightarrow 1.38\times10^{-23}\times T=\frac{1}{2}\times2.76\times10^{-46}\times25\times10^{24}$

$\Rightarrow T=250\ K$

Q12

$(U_1)_i+(U_2)_i=(U_1)_f+(U_2)_f$

[Internal energy of the system remains conserved]

$\Rightarrow\frac{5}{2}\times3\times R\times250+\frac{3}{2}\times5\times R\times470$

$=\frac{5}{2}\times3\times R\times T+\frac{3}{2}\times5\times R\times T$

$\Rightarrow T=\frac{250+470}{2}=360K$

$\Rightarrow 2 \times \frac{1}{2} \times KT = \frac{1}{2} I \omega_f^2$

$\Rightarrow 1.38 \times 10^{-23} \times 360 = \frac{1}{2} \times 2.76 \times 10^{-46} \times \omega_f^2$

$\Rightarrow \omega_f = 6 \times 10^{12}$ rad/s

Q13

For equal PE

$m_A g 4h = m_B g h$

$4m_A = m_B$

For net torque to be zero

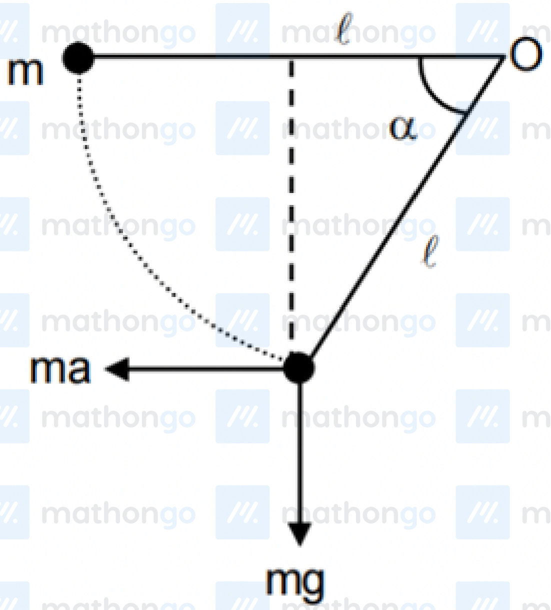
$m_A g(0.75) + 1(0.25) = m_B(0.25)$

$3m_A g + 1 = 4m_A g$

$\mathrm{m_A g} = 1$

$xy = 1 + 1 + 4 = 6N$

Q14



Applying W.E.T with respect to O.

$\mathrm{mg}\ell \sin\alpha = \mathrm{ma}\,\ell(1 - \cos\alpha)$

$\frac{2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}{2\sin^2\frac{\alpha}{2}} = \frac{a}{g}; \cot\frac{\alpha}{2} = \frac{4}{3}; \tan\frac{\alpha}{2} = \frac{3}{4}$

the maximum vertical displacement

$= \ell\sin\alpha = \frac{24\ell}{25} = \frac{48}{25}; \ell = 2\text{ m}$

Q15

$P = \frac{dQ}{dt} = C\frac{dT}{dt} = \frac{CT_0(1+a(t-t_0))^{-3/4}}{4}$

$C = \frac{4P}{aT_0(1+a(t-t_0))^{-3/4}}$

$C = \frac{4P}{aT_0^4}T^3$

Q16

$v_0 = \sqrt{\frac{GM}{a}}$

$mv_0a = mvr$

$\frac{1}{2}m\left(v_0^2 + \frac{v_0^2}{4}\right) - \frac{GMm}{a} = \frac{1}{2}mv^2 - \frac{GMm}{r}$

Solving, $\frac{r_{\max}}{r_{\min}} = \frac{2a}{2a/3} = 3$

Q17

Let the rod be displaced by a small angleθ from the equilibrium position. The torque of the restoring spring force

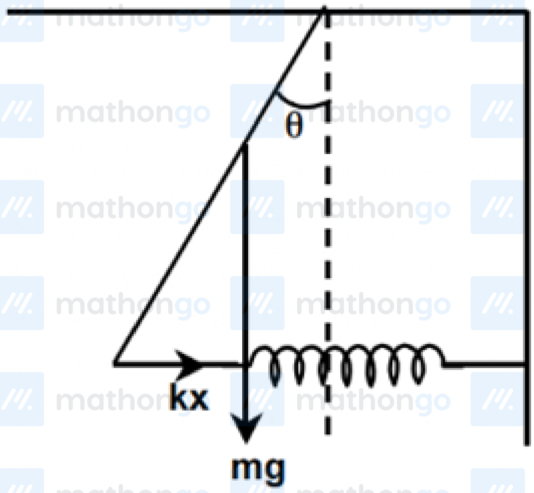
$\tau_1 = K(L\theta)L\cos\theta = K^2\theta$

$[\cos\theta \cong 1]$

The torque of the weight

$\tau_2 = \text{mg}\,\frac{L}{2}\sin\theta = \text{mg}\,\frac{L}{2}\theta[\sin\theta \approx \theta]$

$\therefore$ Total torque $\tau = \tau_1 + \tau_2 = \left(\text{KL}^2 + \frac{\text{mgL}}{2}\right)\theta$



Since torque is against angular displacement

$\alpha = -\left(KL^2 + \frac{mgL}{2}\right)\theta$

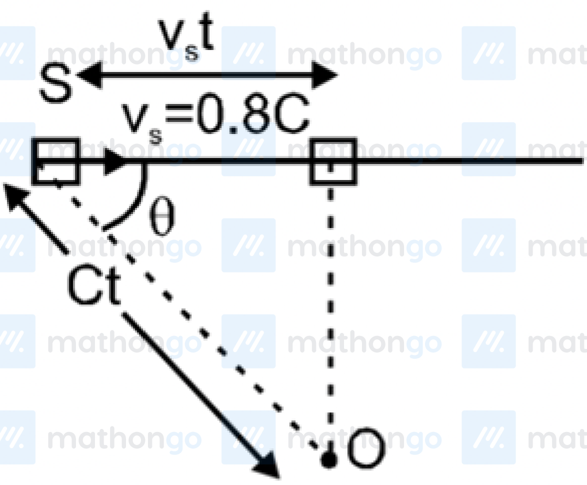
$\therefore \frac{d^2\theta}{dt^2} = -\left(\frac{KL^2}{1} + \frac{mgL}{2l}\right)\theta$

$\therefore \omega = \left(\frac{3K}{m} + \frac{3g}{2L}\right)^{1/2} = \left[3\left(\frac{K}{m} + \frac{g}{2L}\right)\right]^{1/2}$

$\therefore$ Frequency $(f) = \frac{1}{2\pi}\sqrt{3\left(\frac{K}{m} + \frac{g}{2L}\right)}$

Q18

$\cos\theta = \frac{v_s t}{Ct} = 0.8$



$f' = f\left[\frac{v_s t}{C - v_s\cos\theta}\right] = 1.8\left[\frac{C}{C - 0.8C\cos\theta}\right]$

$= 1.8\frac{1}{1-0.8^2} = 1.8\frac{1}{1-0.64} = \frac{1.8}{0.36}$

$= \frac{180}{36} = 5\text{kHz}$

Q19

A is MnO_2. B is HCl. C is Cl_2

$MnO_2 + 4HCl \rightarrow 2H_2O + MnCl_2 + Cl_2$ (Greenish yellow gas C)

Cl_2 (excess) $+NH_3 \rightarrow NCl_3$ (unstable trihalide) $+HCl(B)$

$Cl_2 + NH_3$ (excess) $\rightarrow N_2 + NH_4Cl$

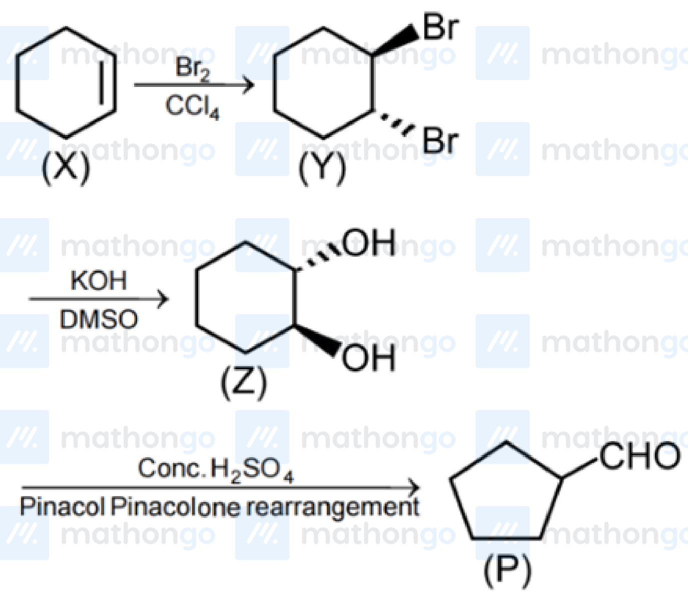
Q20

The compounds are ClF_3, BrF_5 and IF_7. IF_7 contains 21 lone pairs, 3 on each F atom. BrF_5 hybridization is $sp^3\ d^2$. ClF_3 is planar and is used in enrichment of ^{235}U.

$U_{(s)} + 3ClF_{3(l)} \rightarrow UF_{6(g)} + 3ClF(g)$

Q21

P is formed by pinacol-pinacolone rearrangement of vicinal diol Z.



The above product P is formed due to ring contraction during rearrangement. Cyclohexanone is also formed without ring contraction during rearrangement.

Q22

$3.42 \mathrm{ppm} \mathrm{Al}(\mathrm{SO}_4)_3 = \frac{96 \times 3 \times 3.42}{342} \mathrm{ppm}$[SO42-] = 2.88 ppm [SO42-] = 0.54 ppm [Al3+]

1.42 ppm [Na2SO4] = 96/42 x 1.42 ppm [SO42-] = 0.96 ppm [SO42-]

$= \frac{46 \times 1.42}{142} \mathrm{ppm} \mathrm{Na}^+ = 0.46 \mathrm{ppm} \mathrm{Na}^+$

$\left[\mathrm{Al}^{3+}\right] = \frac{0.54 \times 10^3}{27 \times 10^6} = 2 \times 10^{-5} \mathrm{M}, \left[\mathrm{SO}_4^{2-}\right]$

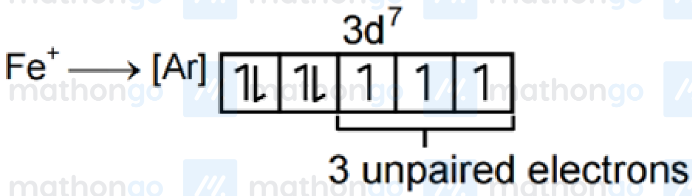
$[\mathrm{SO42\text{-}}] = \frac{(2.88+0.96) \times 10^3}{96 \times 10^6} = 4 \times 10^{-5} \mathrm{M}$

$[\mathrm{Na}^+] = \frac{0.46 \times 10^3}{23 \times 10^6} = 2 \times 10^{-5} \mathrm{M}$

Q23

(A)Fe^{2+} changes to brown-coloured ring complex by charge transfer.

(B)$\mathrm{NO} \rightarrow \mathrm{NO}^+ + \mathrm{e}^- \quad \mathrm{Fe}^{2+} + \mathrm{e}^- \rightarrow \mathrm{Fe}^+$



(C)

Magnetic moment $= \sqrt{n(n+2)}$

$= \sqrt{3 \times 5} = \sqrt{15} = 3.87$ B.M.

$(D) \rightarrow \ sp^3 d^2$ hybridisation

Q24

(A)$\Delta \mathrm{H} = [2(-1263)] - [(-2238) + (-285)] = -3 \mathrm{~kJ}$

(B)$\mathrm{C}_{12}\mathrm{H}_{22}\mathrm{O}_{11} + 12\mathrm{O}_2 \rightarrow 12\mathrm{CO}_2 + 11\mathrm{H}_2\mathrm{O}$

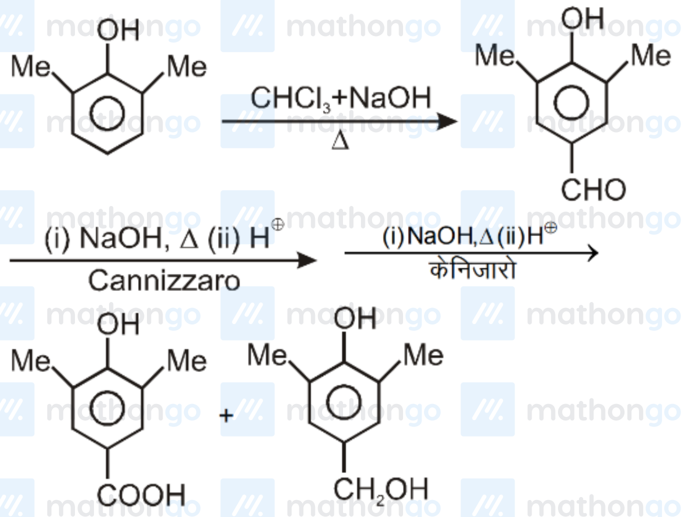
$\{\mathrm{C}_6\mathrm{H}_{12}\mathrm{O}_6 + 6\mathrm{O}_2 \rightarrow 6\mathrm{CO}_2 + 6\mathrm{H}_2\mathrm{O}\} \times 2$

$\mathrm{C}_{12}\mathrm{H}_{22}\mathrm{O}_{11} + \mathrm{H}_2\mathrm{O} \rightarrow 2\mathrm{C}_6\mathrm{H}_{12}\mathrm{O}_6; \Delta \mathrm{H} = -3 \mathrm{~kJ}$

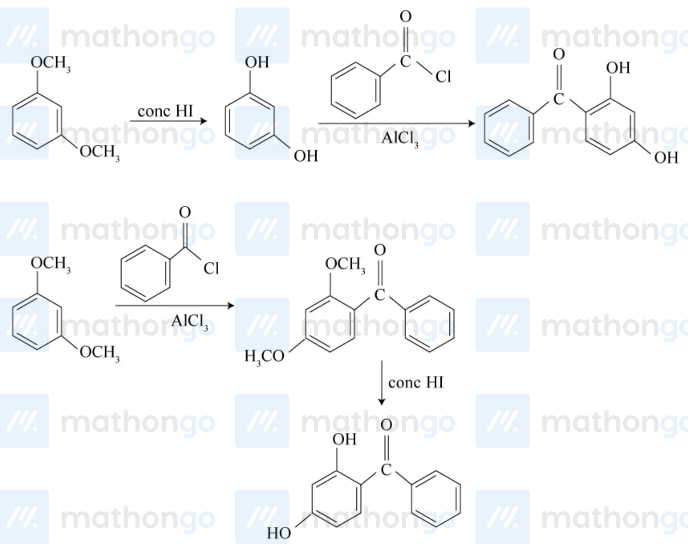
(C)Hydrolysis is exothermic $T \uparrow h \downarrow$

(D)Concentration becomes one-fourth in equal time interval. Hence, 1^{st} order.

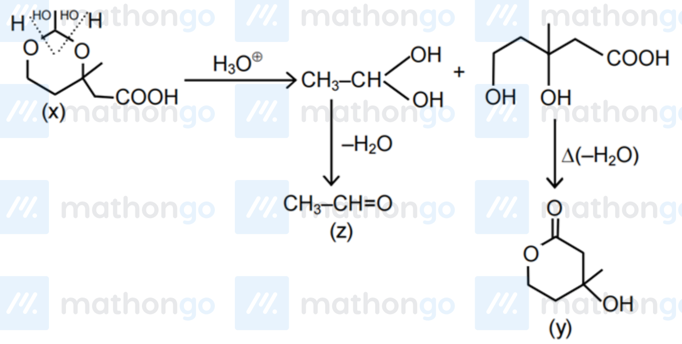
Q25



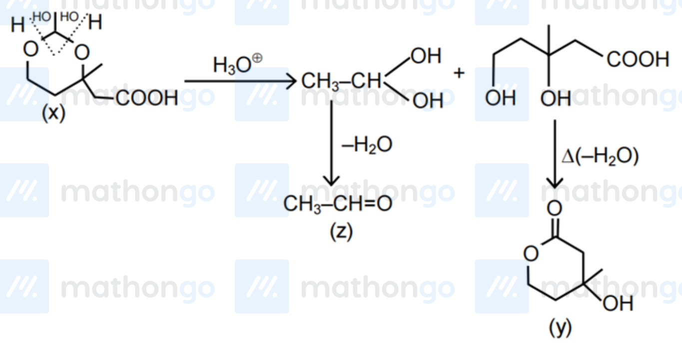
Q26



Q27



Q28



Q29

	$HA(aq.)$	$+$	$NaOH(aq.)$	$\rightarrow$	$NaA(aq.)$	$+$	$H_2O(\ell)$
m $moles$	10		2.5		0		0
m $moles$	7.5		–		2.5		2.5

$\therefore$Resulting solution HA(aq.) and NaA(aq.) is buffer solution.

$$\mathrm{pH} = \mathrm{pK_a} + \log\left(\frac{2.5}{7.5}\right) = 6 + \log\left(\frac{1}{3}\right) = 5.52.$$

Q30

	$HA(aq.)$	$+$	$NaOH(aq.)$	$\rightarrow$	$NaA(aq.)$	$+$	$H_2O(\ell)$
m $moles$	10		10		0		0
m $moles$	0		0		$10\ m\ moles$	of	$NaA(aq.)$

$$\therefore \mathrm{pH} = \frac{1}{2}\mathrm{pK_w} + \frac{1}{2}(\mathrm{pK_a} + \log \mathrm{C}) = 7 + \frac{1}{2}\left[6 + \log\left(\frac{10}{200}\right)\right]$$

$$= 9.35$$

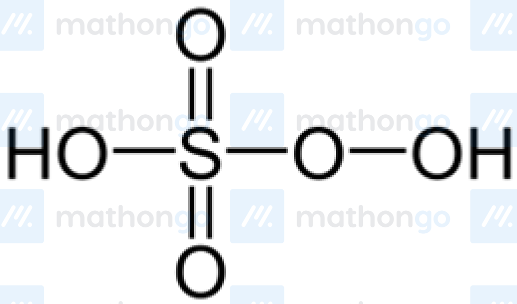
Q31

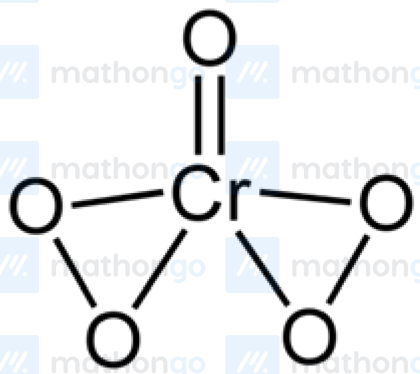
$0.93 = i \times 1.86 \times 0.25 \Rightarrow i = 2$

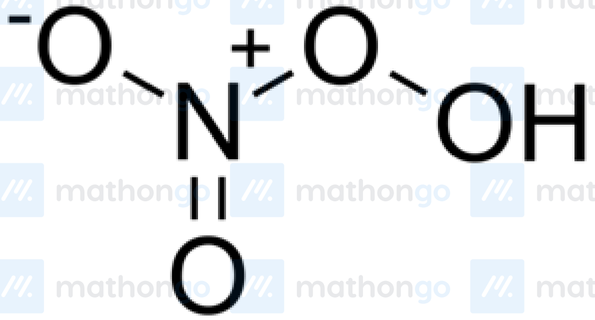
All ammonia ligands must be inside coordination sphere, hence, formula of the complex is $\left[\mathrm{Co}\left(\mathrm{NH}_3\right)_4 \mathrm{Cl}_2\right] \mathrm{Cl}$. Which has two geometrical isomers: cis and trans; both of which are achiral.

Q32

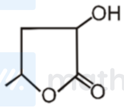
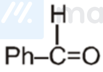
3(d, f, c)

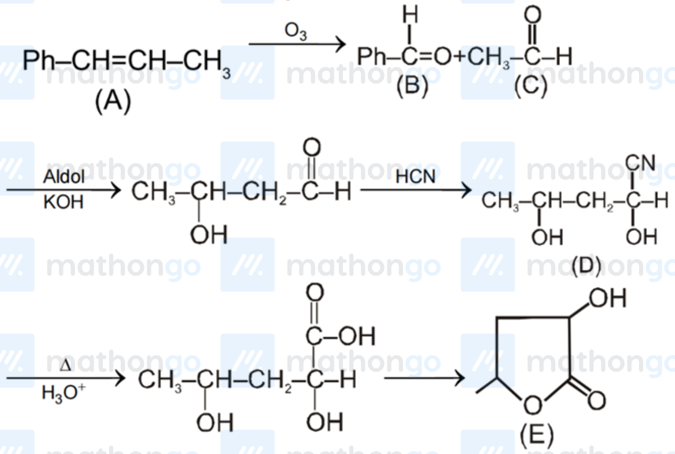




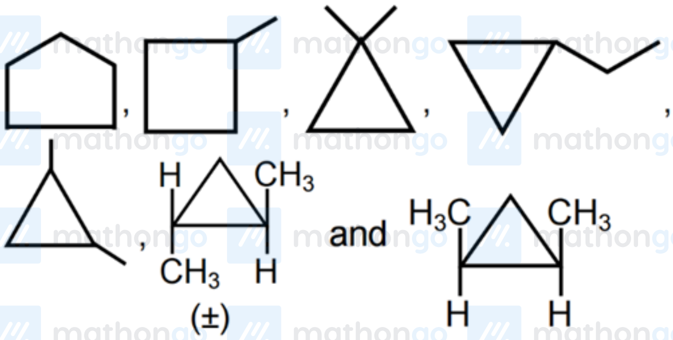


Q33

The structure of E is  and B is 



Q34



Q35

Process $AC =$ polytropic process $(P = KV)$

Molar Heat capacity $\mathrm{C_m = C_V + R/2 = 2R}$

Process $\mathrm{AB = Isobaric}$

$c_m = c_p = 5R/2$

$$\frac{q_{AC}}{q_{AB}} = \frac{\int_{T_A}^{T_C} nC_m \cdot dT}{T_B n \cdot C_{p,m^*} dT} = \frac{2R}{\frac{5}{2}R} = 0.8$$

$$\frac{\mathrm{q_{AC}}}{\mathrm{q_{AB}}} = \times 10 = 0.8 \times 10 = 8$$

Q36

$\Delta U = \text{q} + \text{w}$

For adiabatic process $q = 0$

$\Delta U = w$

$\Delta U = P\Delta V = -P(V_2 - V_1)$

$\Delta U = -100(99 - 100) = 100$ bar mL

Now $\Delta H = \Delta U + \Delta(PV)$

$\Delta H = 100 + [100 \times 99 - 1 \times 100]$

$\Delta H = 9900$ bar mL

Q37

$\alpha = \omega + 3i, \beta = \omega + 9i, \gamma = 2\omega - 4$

Let $\omega = x + iy$

$\therefore \alpha = x + i(y + 3), \beta = x + i(y + 9)$

$\gamma = (2x - 4) + i(2y)$

$\alpha + \beta + \gamma = -a$

$\Rightarrow 4x - 4 + i(y + 3 + y + 9 + 2y) = -a$

$\because$ a is real $\Rightarrow 4y + 12 = 0 \Rightarrow y = -3$

$\therefore \alpha = x, \beta = x + 6i, \gamma = 2x - 4 - 6i$

$\alpha \cdot \beta \cdot \gamma = -c$

$\Rightarrow x[x + 6i)(2x - 4 - 6i) = -c$

$\Rightarrow x[x(2x - 4) + 36 + i\{(2x - 4)6 - 6x\}] = -c$

$\therefore c$ is real

$\Rightarrow x[(2x - 4)6 - 6x] = 0$

$\Rightarrow x = 4$

$\therefore \alpha = 4, \beta = 4 + 6i, \gamma = 4 - 6i$

$a = -12$

$b = 84$

$c=-208$

$|a+b+c|=136$

Q38

$x^4-n\cdot x\cdot+63=(x-\alpha)\left(x^3+ax^2+bx+c\right)$

$\because$ Product of roots $=63$

from (i) $\Rightarrow \alpha$ can be 1,3,7,9 or 63

$x^4-nx+63=x^4+(a-\alpha)x^3+(b-a\alpha)x^2+(c-\alpha b)x-c\alpha$

$\Rightarrow \mathrm{a}-\alpha=0,\ \mathrm{b}-\mathrm{a}\alpha=0, \mathrm{c}-\mathrm{b}\alpha=-\mathrm{n}$

and $-c\alpha=63$

$\Rightarrow \mathrm{n}=\alpha^3+\frac{63}{\alpha}$

when $\alpha=1, \quad \mathrm{n}=64$

$\alpha=3\Rightarrow \mathrm{n}=3^3+\frac{63}{3}=27+21=48$

$\alpha=7\Rightarrow \mathrm{n}=7^3+9$

$\alpha=9\Rightarrow \mathrm{n}=9^3+7$

$\therefore$ least value of n is 48

$\mathrm{n}=48=2^4 3^1$

$\therefore$ number of divisors $=(4+1)(1+1)=10$

number of odd divisors $=2$

$x\cdot y=n=2^4\cdot 3^1$

let $x=2^{\alpha_1}3^{\beta_1}$

$y=2^{\alpha_2}3^{\beta_2}$

$xy=48\Rightarrow 2^{\alpha_1+\alpha_2}\cdot 3^{\beta_1+\beta_2}=2^4\cdot 3^1$

$\Rightarrow \quad \alpha_1+\alpha_2=4 \quad 4+2-1C_4=4$

$\beta_1+\beta_2=1 \quad {}^{1+2-1}C_1=2$

$\therefore$ number of natural solution of $xy=48$ is 10

$\int_0^n e^{\{x\}}dx=\int_0^{48} e^{\{x\}}dx$

$=48\int_0^1 e^x dx=48(e-1)$

Q39

Given A $(a, 0)$

Let $A'(R\cos\alpha, R\sin\alpha)$

Let $P(x, y)$ be any point on M N then

$|PA'| = |PA|$

$\Rightarrow (\mathrm{x} - \mathrm{R}\cos\alpha)^2 + (\mathrm{y} - \mathrm{R}\sin\alpha)^2 = (\mathrm{x} - \mathrm{a})^2 + \mathrm{y}^2$

$\Rightarrow x\cos\alpha + y\sin\alpha = \frac{R^2 - a^2 + 2ax}{2R}$

$\Rightarrow \frac{x\cos\alpha + y\sin\alpha}{\sqrt{x^2+y^2}} = \frac{R^2 - a^2 + 2ax}{2R\sqrt{x^2+y^2}}$

$\Rightarrow \sin(\theta + \alpha) = \frac{\mathrm{R}^2 - \mathrm{a}^2 + 2\mathrm{aax}}{2\mathrm{R}\sqrt{\mathrm{x}^2+\mathrm{y}^2}}$

where $\sin\theta = \frac{\mathrm{x}}{\sqrt{\mathrm{x}^2+\mathrm{y}^2}}, \cos\theta = \frac{\mathrm{y}}{\sqrt{\mathrm{x}^2+\mathrm{y}^2}}$

$\because |\sin(\theta + \alpha)| \leq 1$

$\Rightarrow \left|\frac{R^2 - a^2 + 2ax}{2R\sqrt{x^2+y^2}}\right| \leq 1$

$\Rightarrow$ centre $\left(\frac{a}{2}, 0\right), e = \sqrt{1 - \frac{\frac{R^2 - a^2}{4}}{\frac{R^2}{4}}} = \frac{a}{R}$

$\Rightarrow \frac{(2x-a)^2}{R^2} + \frac{4y^2}{R^2 - a^2} \geq 1$

Q40

$\because f(xy+1) = f(x)\cdot f(y) - f(y) - x + 2 \quad \forall x, y \in R$

$\therefore f(yx+1) = f(y)\cdot f(x) - f(x) - y + 2$

$\Rightarrow f(x)\cdot f(y) - f(y) - x + 2 = f(y)\cdot f(x) - f(x) - y + 2$

$\Rightarrow f(x) + y = f(y) + x$

put $y = 0$

$f(x) + 0 = f(0) + x = 1 + x$

$\Rightarrow f(x) = x + 1$

$\sin^{-1}\left(\sin\left(f^{-1}(7)\right) = \sin^{-1}(\sin 6) = 6 - 2\pi\right.$

10

$$\sum_{k=0} f(k) = \sum_{k=0}^{10}(k+1) = 1 + 2 + 3 \ldots . +1\Big) = \frac{11\times 12}{2}$$

$\therefore f(yx+1) = f(y)\cdot f(x) - f(x) - y + 2$

$\Rightarrow f(x)\cdot f(y) - f(y) - x + 2 = f(y)\cdot f(x) - f(x) - y + 2$

$\Rightarrow f(x) + y = f(y) + x$

put $y = 0$

$f(x) + 0 = f(0) + x = 1 + x$

$\Rightarrow f(x) = x + 1$

$\sin^{-1}\left(\sin\left(f^{-1}(7)\right) = \sin^{-1}(\sin 6) = 6 - 2\pi\right.$

$\sum\limits_{k=0}^{10} f(k) = \sum\limits_{k=0}^{10}\left.(k+1) = 1 + 2 + 3 \ldots . + 1\right) = \frac{11\times12}{2} = 66$

$f(2016) = 2017$ is prime

Q41

$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \ldots\ldots\ldots$

$= \frac{1}{2}\begin{bmatrix} e^{2x}+1 & e^{2x}-1 \\ e^{2x}-1 & e^{2x}+1 \end{bmatrix}$

$\therefore f(x) = e^{2x} + 1, g(x) = e^{2x} - 1$

$\int_0^1 \left(f\left(\frac{x^2}{2}\right) - 1\right)(x-\alpha)dx = 0$

$\Rightarrow \int_0^1 e^{x^2}(x-\alpha)dx = 0$

$\Rightarrow \int_0^1 e^{x^2}x \quad dx = \int_0^1 e^{x^2}\alpha dx$

$\Rightarrow \frac{1}{2}(e-1) = \alpha\int_0^1 e^{x^2}dx \Rightarrow \alpha = \frac{\frac{1}{2}(e-1)}{\int_0^1 e^{x^2}dx}$

$\because 0 < \alpha < 1 \Rightarrow 1 < \int_0^1 \mathrm{e}^{\mathrm{x}^2}\mathrm{dx} < \mathrm{e}$

$f(x)dy + y\cdot g(x)dx = 0$

$\Rightarrow \frac{dy}{y} = -\frac{g(x)}{f(x)}dx. \Rightarrow \frac{dy}{y} = -\frac{(e^{2x}-1)}{e^{2x}+1}dx$

integrating both sides

$\ell n\, y = -\int \frac{e^{2x}-1}{e^{2x}+1}dx$

$= -\int \frac{e^x-e^{-x}}{e^x+e^{-x}}dx = -\ln(e^x + e^{-x}) + \ell nc$

$\Rightarrow y = \frac{c}{e^x+e^{-x}} = \frac{c\cdot e^x}{e^{2x}+1} = \frac{c\cdot e^x}{f(x)}$

Q42

Let $P = (1+x)\left(1+x+x^2\right)\left(1+x+x^2+x^3\right)\ldots\ldots.(1+x+\ldots.x^n)$ term containing highest power of x in P is

$x\cdot x^2\cdot x^3\ldots\ldots.x^n = x^{\frac{n(n+1)}{2}}$

$\therefore$ total number of terms in the product

$= \frac{n(n+1)}{2} + 1 = \frac{n^2+n+2}{2}$

Let $m = \frac{n+(n+1)}{2}$ then

$P = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots\ldots a_mx^m$

$\Rightarrow (1+x)\left(1+x+x^2\right)\ldots\ldots\left(1+x+x^2+x^3+\ldots.+x^n\right)$

$= a_0 + a_1x + a_2x^2 + \ldots + a_mx^m$

$\Rightarrow (1+x)\left(1+x+x^2\right)\ldots\ldots\left(1+x+x^2+x^3+\ldots.+x^n\right)$

$= a_0 + a_1x + a_2x^2 + \ldots + a_mx^m$

Replacing x by $\frac{1}{x}$, we get

$\left(1+\frac{1}{x}\right)\left(1+\frac{1}{x}+\frac{1}{x^2}\right)\ldots.\left(1+\frac{1}{x}+\frac{1}{x^2}+\ldots+\frac{1}{x^n}\right)$

$= a_0 + a_1 \cdot \frac{1}{x} + a_2 \cdot \frac{1}{x^2} + \ldots.\ a_m \cdot \frac{1}{x^m}$

$\Rightarrow (1+x)\left(1+x+x^2\right)\ldots.(1+x+\ldots x^n) = a_0x^m + a_1x^{m-1} + \ldots.\ a_m$

from (i) and (ii)

$a_0 = a_m,\ a_1 = a_{m-1},\ \ldots,\ a_m = a_0$

$\Rightarrow$ coefficient of terms equidistant from beginning and end are equal

put $x = 1$ and $x = -1$ in (i), we get

$a_0 + a_1 + a_2 + \ldots\ldots.a_m = (n+1)!$ .... (iii)

and $a_0 - a_1 + a_2 \ldots..+(-1)^m a_m = 0$......... (iv)

(iii) and (iv) $\Rightarrow a_0 + a_2 + a_4 + \ldots\ldots = \frac{(n+1)!}{2}$

(iii) and (iv) $a_1 + a_3 + a_5 + \ldots.. = \frac{(n+1)!}{2}$

Q43

$P^2 = -2I \Rightarrow |P|^2 = -8$

and $|2R+2I| = 32 \Rightarrow 2^3|R+I| = 32 \Rightarrow |R+I| = 4$

Now;

$P^5 - 2P^3R + QP^2 - 2QR = 0$

$\Rightarrow 4P + 4PR - 2Q - 2QR = 0$

$\Rightarrow 4P(I+R) - 2Q(I+R) = 0$

$\Rightarrow (I+R)(4P-2Q) = 0$

$\Rightarrow 4P - 2Q = 0 \quad (\because I + R$ is non singular)

$\Rightarrow 2P - Q = 0$

$\Rightarrow Q - P = P \Rightarrow |Q - P|^2 = |P|^2 = -8$

Q44

(A)$\because \ \frac{1}{1-\alpha}\int\limits_{\alpha}^{1} g(x)\,dx \leq \frac{1}{\alpha}\int\limits_{0}^{\alpha} g(x)\,dx$

$\alpha\int\limits_{\alpha}^{1} g(x)\,dx \leq (1-\alpha)\int\limits_{0}^{\alpha} g(x)\,dx$

Add $\int_0^\alpha f(x)dx$ both sides

$\alpha\int_0^1 f(x)dx \leq \int_0^\alpha f(x)dx$

(B)Cauchy-schwarz inequality to the functions $f(t)$ and $g(t) = 1 \forall t \in [0,1]$

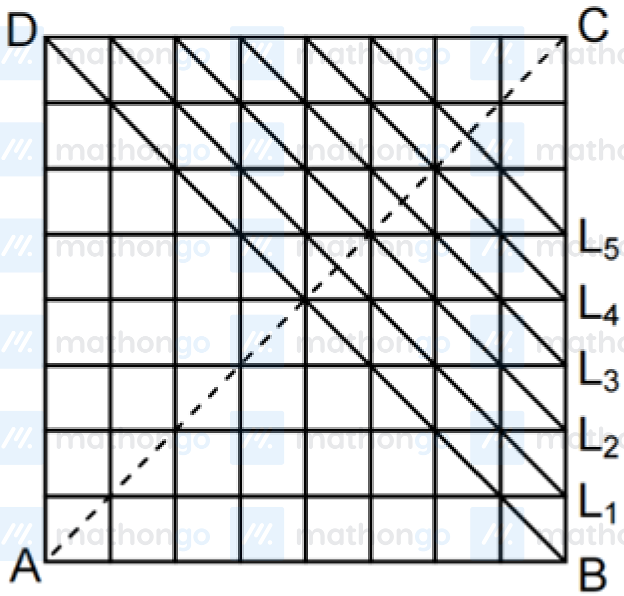
Q45

Let A be the event that squares have a side in common $\mathrm{n(s)} = {}^{64}\mathrm{C}_2; \mathrm{n(A)} = 7.8 + 7.8 = 112$

$P(A) = \frac{n(A)}{n(S)} = \frac{1}{18}$

reqd. prob $= 1 - P(A) = 1 - \frac{1}{18} = \frac{17}{18}$

Q46



$\mathrm{n(S)} = {}^{64}\mathrm{C}_3 = 41664$

total no. of favourable cases

$n(A) = 2 \cdot {}^{8}C_3 + 4\left[{}^{7}C_3 + {}^{6}C_3 + {}^{5}C_3 + {}^{4}C_3 + {}^{3}C_3\right]$

Required. prob $= \frac{392}{41664} = \frac{7}{744}$

Q47

$F'(x) = e^{(1+\sin^{-1}(\cos x))^2} \cdot (-\sin x) - e^{(1+x)^2} \cdot \cos x$

$F'(0) = -e = F'\left(\frac{\pi}{2}\right)$

Since $F'(x)$ is continuous and differentiable and

$F'(0) = F'\left(\frac{\pi}{2}\right)$

Hence Rolle's theorem is applicable for $y = F'(x)$ in $\left[0, \frac{\pi}{2}\right]$

$\therefore \exists$ some $c \in \left(0, \frac{\pi}{2}\right)$ s.t. $F''(c) = 0$

Q48

Differentiating the equation w.r. to x, we get $2f(x)f'(x) = (f(x))^2 + (f'(x))^2$

$\Rightarrow (f(x) - f'(x))^2 = 0 \Rightarrow f(x) = f(x)$

$\Rightarrow \frac{f'(x)}{f(x)} = 1 \Rightarrow \ln f(x) = x + c$

but $x = 0, f(0) = \sqrt{m^2} \quad m \in R^+$

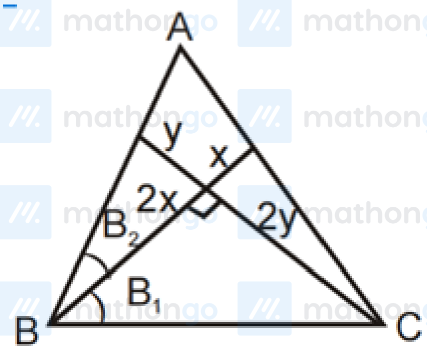
$\therefore \mathrm{c} = \ln \mathrm{m}$

$\therefore \ \ln f(x) = x + \ln m \Rightarrow \ln\left(\frac{f(x)}{m}\right) = x$

$\Rightarrow f(x) = me^x, \ m \in R^+$

$f'(x) > 0 \,\forall\, x \in R$

Q49



$\tan B_1 = \frac{2y}{2x}, \tan B_2 = \frac{y}{2x}$

$B = B_1 + B_2$

$\tan B = \frac{\frac{y}{x}+\frac{y}{2x}}{1-\frac{y}{x}\frac{y}{2x}} = \frac{3yx}{2x^2-y^2}$

$\cot B = \frac{2x^2-y^2}{3xy}$

Similarly $\cot C = \frac{2y^2-x^2}{3xy}$

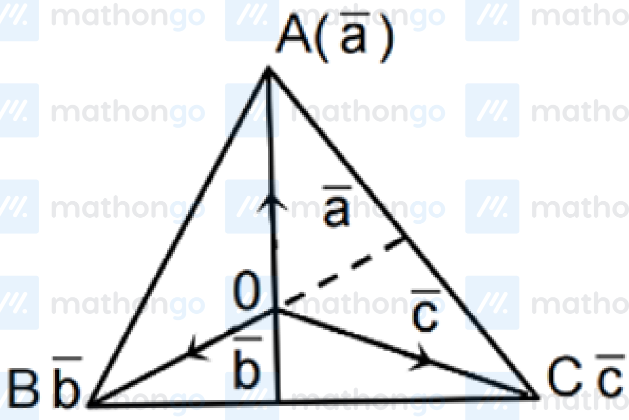
$\cot B + \cot C = \frac{x^2+y^2}{3xy} \geq \frac{2}{3}$

Q50

Let O be origin and position vector of A, B, C are $\vec{a}, \vec{b}, \vec{c}$ respectively

$\because\ \overrightarrow{OA} + 2\overrightarrow{OB} + 3\overrightarrow{OC} = \vec{0}$

$\Rightarrow\ \vec{a} + 2\vec{b} + 3\vec{c} = \vec{0}$ (1)



Now area of $\Delta ABC = \frac{1}{2}|\vec{b}\times\vec{c} + \vec{c}\times\vec{a} + \vec{a}\times\vec{b}|$

from equation (1) $\vec{a} \times \vec{b} + 2\vec{b} \times \vec{b} + 3\vec{c} \times \vec{b} = \overrightarrow{0}$

& $\vec{a} \times \vec{c} + 2\vec{b} \times \vec{c} + 3\vec{c} \times \vec{c} = \vec{0}$

$\vec{a} \times \vec{b} = -3\vec{c} \times \vec{b} = 3\vec{b} \times \vec{c}$

$\vec{a} \times \vec{c} + 2\vec{b} \times \vec{c} + 3\vec{c} \times \vec{c} = \overrightarrow{0}$

$\Rightarrow \vec{b} \times \vec{c} = \dfrac{-\vec{a} \times \vec{c}}{2} = \dfrac{\vec{c} \times \vec{a}}{2}$

$\Rightarrow \vec{b} \times \vec{c} = \frac{-\vec{a} \times \vec{c}}{2} = \frac{\vec{c} \times \vec{a}}{2}$

$\Rightarrow$ Area of $\Delta\mathrm{ABC} = 3 \cdot \frac{1}{2}|\overrightarrow{\mathrm{C}} \times \overrightarrow{\mathrm{a}}| = 3 \cdot \Delta\mathrm{AOC}$

Q51

Let $[\sqrt{n}] = [\sqrt{n+125}] = k$

$\Rightarrow k^2 \leq n < n + 125 < (k+1)^2$

$\Rightarrow \mathrm{n} + 125 < \mathrm{k}^2 + 2\mathrm{k} + 1 \leq \mathrm{n} + 2\mathrm{k} + 1$

$\Rightarrow 2\mathrm{k} + 1 > 125 \Rightarrow \mathrm{k} > 62$

$\because \mathrm{k}^2 \leq 5000 \Rightarrow \mathrm{k} \leq 70$

$\Rightarrow \mathrm{k} \in \{63, 64, 65, 66, 67, 68, 69, 70\}$

$\because 63^2 = 3969$ and $64^2 = 63^2 + 127$

$\therefore \left[\sqrt{63^2 + 125}\right] = \left[\sqrt{63^2 + 1 + 125}\right] = 63$

$\Rightarrow$ two values of m

if $\mathrm{k} = 64$

$\left[\sqrt{64^2 + 125}\right] = \left[\sqrt{64^2 + 126}\right] = \left[\sqrt{64^2 + 127}\right]$

$= \left[\sqrt{64^2 + 128}\right] = 64$

$\Rightarrow$ four values of n

continuing we see that there an 6,8,10,12,14,16 values of n respectively for $\mathrm{k} = 65, 66, 67, 68, 69, 70$

$\therefore$ total values of n

$= 2 + 4 + 6 + 8 + 10 + 12 + 14 + 16 = 72$

$= \mathrm{m} = 72$

sum of digits $= 9$

Q52

$a!+b!+c!=3^d$

if $a>1$ then L.H.S even R.H.S odd

$\Rightarrow a=1$

$\therefore b!+c!=3^d-1$ (2)

if $b>2$ then L.H.S multiple of 3 R.H.S is not multiple of 3

$\Rightarrow b=1$ or $b=2$

of $b=1$ then from (3)

$c!=3^d-2$

$\Rightarrow c<2 \Rightarrow c=1,$

$\Rightarrow d=1$

If $b=2$ then $c!=3^d-3$

$\Rightarrow d=2, c=3$

$d=3, c=4$

$\Rightarrow (a, b, c, d)=(1,1,1,1),(1,2,3,2),(1,2,4,3)$

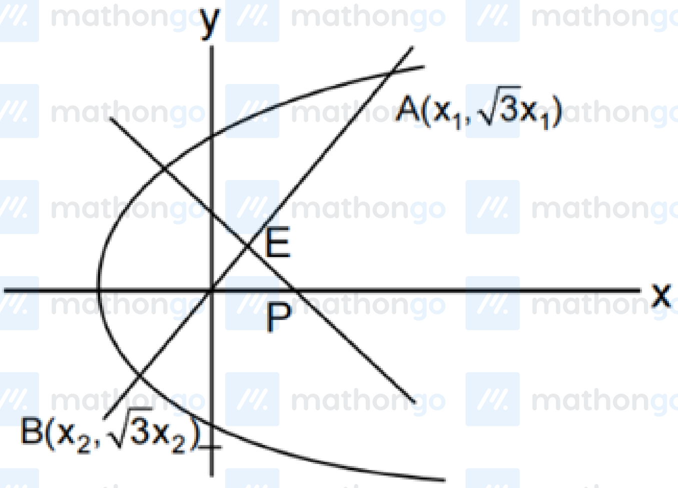
Q53

$F(0,0), Eq^n$ of st. line Through points A and B will be $y=\sqrt{3x}$ Substitute it into the e^n of the parabola. We get $3x^2-8x-16=0$ If roots are x_1 and x_2 then $x_1+x_2=\frac{8}{3}$ and $x_1x_2=-\frac{16}{3}$

Now $A\left(x_1, \sqrt{3}x_1\right)$ and $B\left(x_2, \sqrt{3}x_2\right)$ then mid-point A B is

$E\left(\frac{x_1+x_2}{2}, \sqrt{3}\left(\frac{x_1+x_2}{2}\right)\right)=\left(\frac{4}{3}, \frac{4}{\sqrt{3}}\right)$

Equation of perpendicular bisector is $y-\frac{4}{\sqrt{3}}=-\frac{1}{\sqrt{3}}\left(x-\frac{4}{3}\right)$



$\Rightarrow \sqrt{3}y - 4 = -x + \frac{4}{3}$ Hence P is $\left(\frac{16}{3}, 0\right)$ $\ell = \frac{16}{3} \Rightarrow \frac{3}{2}\ell = 8$

Q54

Let parabola be $y^2 = 4ax$ and P $\left(at_1^2, 2at_2\right)$ and $Q(at_2^2, 2at_2)$, be the point of intersection of tangent at t_1 and t_2

$= T\left(at_1t_2, a\left(t_1 + t_2\right)\right)$

$P' \equiv \left(at_3t_1, a\left(t_3 + t_1\right)\right)$

$Q' \equiv \left(at_2t_3, a\left(t_2 + t_3\right)\right)$

$$\frac{TP'}{TP} = \frac{\lambda}{1} \Rightarrow \lambda = \frac{t_3 - t_2}{t_1 - t_2}$$

$Q' \equiv \left(at_2t_3, a\left(t_2 + t_3\right)\right)$

$\frac{TP'}{TP} = \frac{\lambda}{1} \Rightarrow \lambda = \frac{t_3 - t_2}{t_1 - t_2}$

Similarly $\frac{TQ'}{TQ} = \frac{t_1 - t_3}{t_1 - t_2}$

$\therefore \frac{TP'}{TP} + \frac{TQ'}{TQ} = 1$